

Association of Food Consumption and Child Stunting

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Overall structure of my thesis

- ① Background
- ② Research interest
- ③ Factor analysis of stunting reduction focusing on food consumption (IV: CCT program participation rate)
- ④ Impact of multisector intervention on stunting reduction
- ⑤ 2SLS focusing on interaction
- ⑥ Impact of CRECER program in stunting reduction
- ⑦ PSM-DID for impact assessment of CRECER program
- ⑧ Mediation analysis to assess contribution of CRECER program
- ⑨ Discussion

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- ③ Data and methodology
- ④ Result
- ⑤ Discussion

Section 1

Background

Current status

- Malnutrition is one of the world's most serious but least-addressed development challenges. Its human and economic costs are enormous, falling hardest on the poor, women, and children.
- South Asia and Sub-Saharan Africa are the hardest hit, with with roughly two-thirds of the 150.2 million stunted children residing in these two regions(World bank).
- The economic costs of undernutrition are significant with at least \$1 trillion a year because of productivity loss due to undernutrition and micronutrient deficiencies (FAO 2013).
- From meta-analysis of 37 international study, it is revealed that stunting was strongly linked to household food insecurity and poor dietary diversity (2025, islam et. al).
- Meanwhile, it is well known that other various factors including mother's education status, access to clean water and toilet, economic disparity, fragile agro-ecosystem have also significant impact on malnutrition.

Why we need multisector intervention?

three key factors are associated with undernutrition

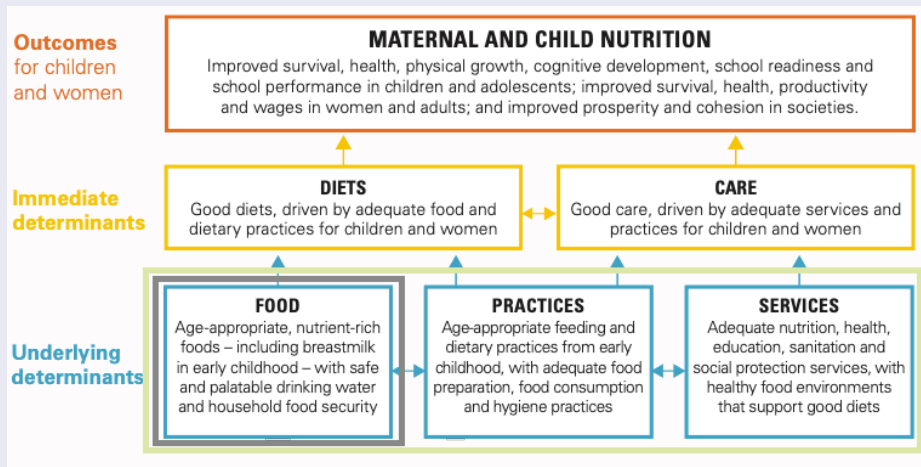


Figure 1: UNICEF framework of undernutrition

Peru case study

Key factors associated with undernutrition in Peru

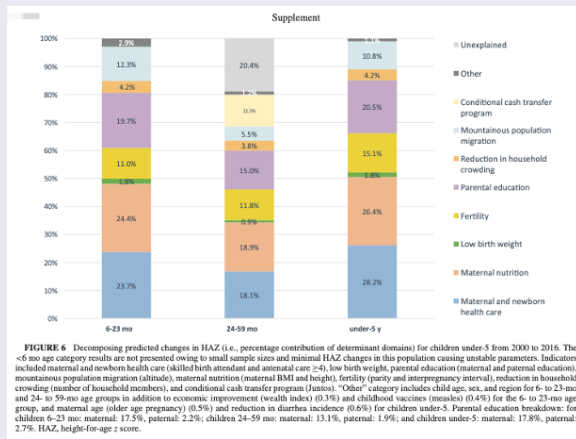


Figure 2: Drivers of stunting reduction in Peru: a country case study (Luis, et. Al, 2020)

Different type of dietary guidance

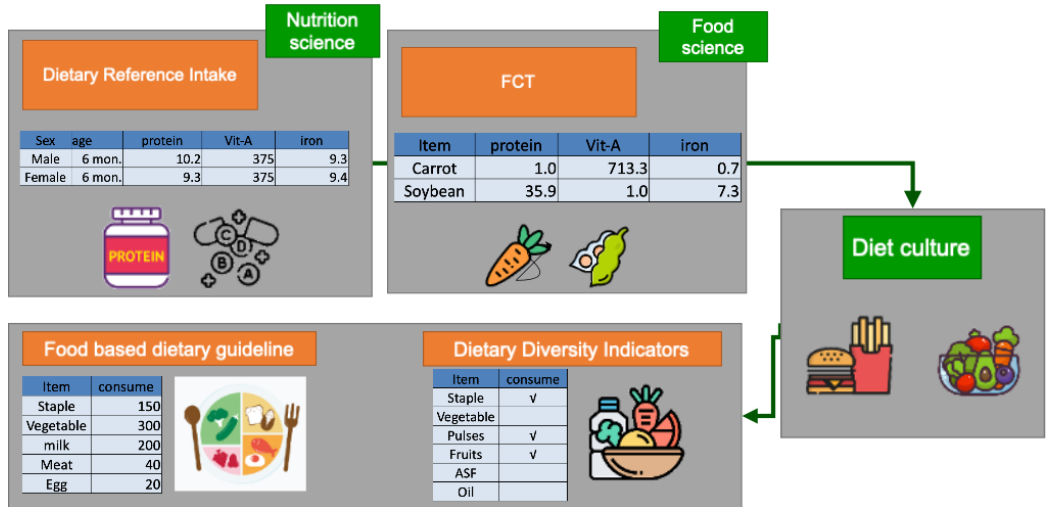


Figure 3: different type of dietary guideline

How to evaluate quality of diet?

Table 1: Key consideration in evaluating quality of diet

Key consideration in evaluating quality of diet	
Diversity	Diversity of food group (qualitative)
Balance	Consumed food group balance (semi-quantitative)
Adequacy	Adequacy of key nutrient intake from diet (quantitative)
Moderation	Refrain overconsumption

Is Dietary Diversity Score effective tool to assess diet quality?

- Ruel conducted intensive review of 47 articles to examine impact of various agriculture intervention to nutrition improvement. Overall, number of studies showed clear impact of NSA to DDIs, while impact on nutrition outcome was limited (Ruel, 2018)
- Eric conducted peer review from 137 articles to examine relationship between DDIs and health impact, found that associations between DDIs and health outcomes were largely inconsistent (Eric, 2021)

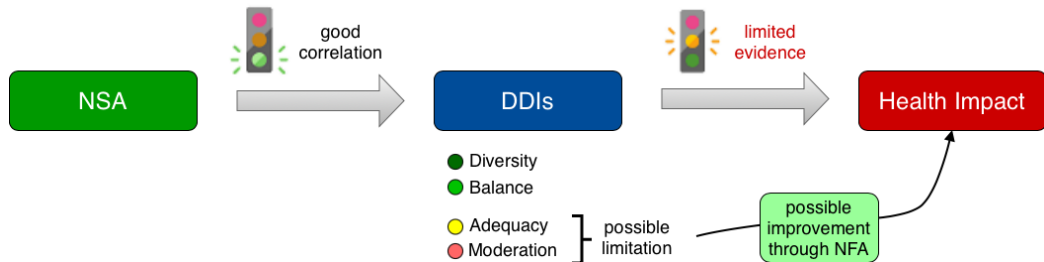


Figure 4: effectiveness of DDIs

Descriptive statistics1

Characteristic	N	control_pre(<2009) N = 21,836 ¹	treat_
Height under 145cm - women	30,017	2,426 (11%)	1
Birth weight less than 2.5 kg	30,017	1,289 (5.9%)	4
more than 3 child under 12	30,017	3,140 (14%)	1
secondary or higher education	30,017	11,414 (52%)	2
own a sheep	30,017	2,950 (14%)	1
Skilled assistance during ANC	30,017	9,196 (42%)	4

¹n (%); Mean (SD)

²Design-based Wald test of association; Design-based t-test

Descriptive statistics1

Characteristic	N	control_pre(<2009) N = 21,836 ¹	t
owns refrigerator	30,017	7,023 (32%)	
Overweight or obese BMI - women	30,017	11,309 (52%)	
water fetching over 30min	30,017	74 (0.3%)	
HH head over 29	30,017	17,560 (80%)	
living in capital city	30,017	3,325 (15%)	
open defecation	30,017	3,301 (15%)	

¹n (%); Mean (SD)

²Design-based Wald test of association; Design-based t-test